

Energy-Efficient Optical Chiplet Link with Heterogeneously Integrated Membrane EA-DFB Laser Achieving 1.55 pJ/bit at 64-Gbit/s PAM-4

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Abstract We demonstrate a transmitter chiplet integrating a low-capacitance InGaAsP electro-absorption modulator and a membrane InGaAsP laser on a Si platform, co-integrated with an electronic IC through flip-chip bonding. Together with a Ge receiver chiplet, a 1.55-pJ/bit optical link at 64-Gbit/s PAM-4 is achieved. ©2026 The Authors

Introduction

Recent growth in artificial intelligence (AI) and high-performance computing systems is driving a strong demand for optical interconnects with high bandwidth density and low energy consumption. Co-packaged optics (CPO) is becoming an important approach for short-reach links around compute packages, where electrical interconnects are constrained by transmission loss, reach, and power. CPO systems also require sufficient tolerance to the package operating environment together with a small footprint and low implementation cost. Optical chiplets based on silicon photonic integrated circuits (PICs) have therefore emerged as a promising solution for dense on-package optical I/O [1].

In many optical chiplets with transmitters (Tx), optical power is supplied from an external laser source [2–6]. This approach is attractive because it allows laser operation at cooler locations away from high-temperature ASICs and facilitates system-level replacement and serviceability [5]. In our previous work [7], we also utilised an external laser source. The PIC included an optical input port and a 1×4 splitter to feed optical power to four-channel electro-absorption modulators (EAMs), as shown in Fig. 1(a).

While this architecture is effective, the injected light must pass through fibre interfaces and passive silicon-photonic circuits, introducing additional optical loss due to coupling, routing, and mode conversion. In addition, when a single light source is shared by multiple EAMs, higher launched optical power is required to compensate for splitting loss, which becomes more pronounced as the channel count increases. Thus, we explore on-chip laser integration as an alternative approach to reduce the optical overhead associated with external light distribution in high-density optical interconnects.

Membrane EAMs have shown a high electro-

optic (E-O) bandwidth of 103 GHz and 400-Gbit/s four-level pulse-amplitude modulation (PAM-4) operation, thanks to the low junction capacitance [8]. In addition, for the driver electronic integrated circuits (EICs), the low junction capacitance enables a high-output-impedance, low-voltage driver design, thereby reducing the power consumption [7]. The next challenge is the integration of an on-chip laser into optical chiplets, as shown in Fig. 1(b), to further reduce the total energy consumption. For process simplification, the EAM and laser share the same active layer. Although this constraint can limit device optimisation, the membrane structure enables strong optical confinement and provides a large coupling coefficient through a surface grating. As a result, low-power laser operation is achieved even with a shared active layer [9].

In this work, we fabricated an optical chiplet incorporating a membrane InGaAsP-based EAM integrated with distributed-feedback (EA-DFB) laser, as shown in Fig. 1(c). We demonstrated 64-Gbit/s PAM-4 operation at an energy cost of 1.20 pJ/bit. Furthermore, by combining the chiplet with a Ge-based receiver (Rx), we demonstrated an optical link with a total energy cost as low as 1.55 pJ/bit.

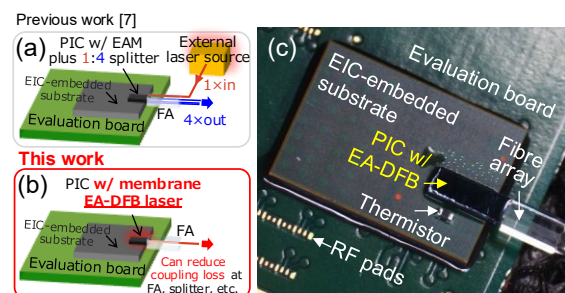


Fig. 1: (a) Schematic diagrams of the Tx chiplet (a) in the previous work [7] and (b) in this work, and (c) the microscopic image of the fabricated Tx chiplet with the membrane EAM integrated with the membrane laser.

Design of PIC and EIC

Figure 2(a) shows a microscope image of the PIC together with a cross-sectional view of the active region. An eight-channel EA-DFB laser array was integrated, and 100- μm -long detuned DFB lasers were designed as internal optical sources. The Bragg wavelengths of the 20- μm -long rear section and the 80- μm -long front section were slightly detuned to enable single-mode operation, and the InP slab thickness was set to 340 nm [10]. The EAM length was set to 100 μm . The DFB and EAM sections consist of the same InGaAsP active region operating in the C-band. The laser and EAM, both having lateral p-i-n structures, were connected by a 250- μm -long InP channel waveguide, which enables electrical isolation. In addition, this separation also provides thermal isolation. A fibre array (FA) with high-numerical-aperture (HNA) fibres was attached to the SiO_x waveguide facet. The HNA fibres were connected to standard single-mode fibres (SSMFs) via thermally expanded core fibres for mode conversion. Loop-back waveguides, shown in Fig. 2(a), were used for FA alignment by maximising the optical power received at the output port.

The CMOS-driver EIC was embedded in the substrate, and the PIC was flip-chip bonded to the EIC through short stacked vias for lumped driving (Fig. 2(b)). This configuration is suitable for high-density integration and enables the EAM to be driven by the EIC without power-consuming termination resistors [7]. The driver EIC bandwidth was designed for 32-GBaud PAM-4 signalling [11,12]. The temperature of the PIC was monitored using a thermistor, as shown in Fig. 1(c), and is hereafter defined as the PIC temperature.

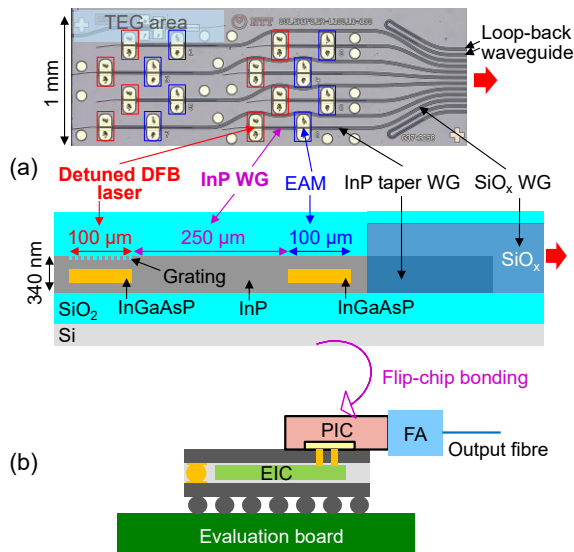


Fig. 2: (a) Microscope image of the PIC and a schematic cross-sectional diagram of the active InGaAsP region in the PIC, including the EAM section and the detuned DFB laser, and (b) schematic diagram of the Tx chiplet.

Experimental Results and Discussion

In this work, we tested single-channel operation. Figure 3(a) shows the fibre output power of the Tx chiplet as a function of laser bias current ($L-I$ curve) at a PIC temperature of 35°C, for EAM bias voltages of 0 and -0.82 V; the latter was used for dynamic modulation. The threshold current was 4 mA. The fibre output power at a laser bias current of 17 mA and an EAM bias voltage of -0.82 V was 0.29 mW, including an optical-switch loss of ~1 dB; the measurement setup is shown later in Fig. 5. Figure 3(b) shows the lasing spectrum at the laser bias current of 17 mA and EAM bias voltage of -0.82 V. We confirmed single-mode lasing from the assembled Tx chiplet.

Next, we discuss the dynamic modulation characteristics. Figure 4 shows the small-signal E-O response of the fabricated chiplet at a laser bias current of 19.5 mA and an EAM bias voltage of -0.82 V, where the RF S21 loss of the evaluation board was de-embedded. The measured 3-dB bandwidth was 24.2 GHz, which is sufficient for 32-GBaud PAM-4.

Figure 5 shows the experimental setup to measure the optical eye diagram of the fabricated Tx chiplet. For large-signal modulation, differential electrical signals with a pseudo-random bit sequence of $2^{11}-1$ were generated by an arbitrary waveform generator (AWG) and applied to the evaluation board through a GSSG probe. The modulated optical signal from the chiplet was detected using a commercial reference Rx (Keysight N1030A), and the electrical waveform was captured with a sampling oscilloscope. The RF loss of the cables and the evaluation board

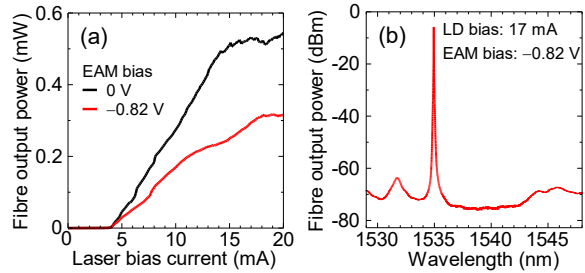


Fig. 3: (a) $L-I$ curves and (b) lasing spectra of the fabricated Tx chiplet at the PIC temperature of ~35°C.

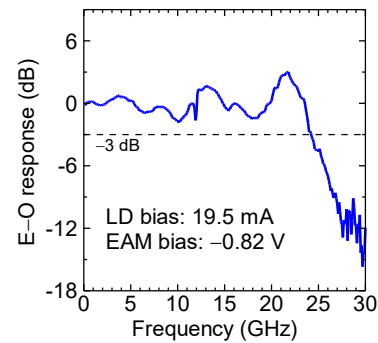


Fig. 4: Small-signal E-O response of the fabricated Tx chiplet.

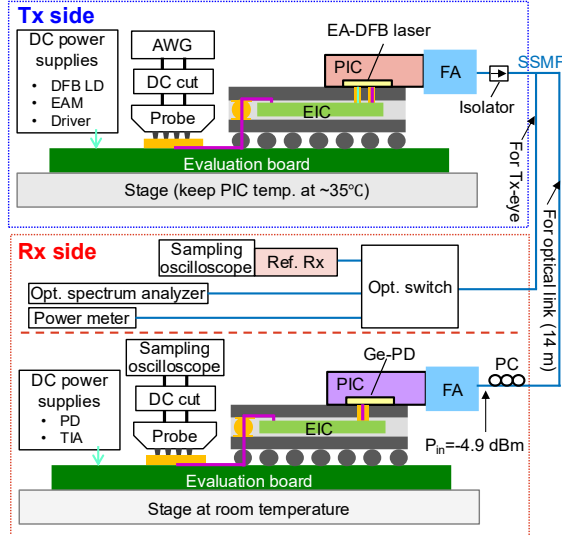


Fig. 5: Experimental setup for measuring the eye diagram of the Tx and optical link.

was compensated for at the AWG side to evaluate the inherent Tx-chiplet performance. We controlled the stage temperature to keep the PIC temperature at $\sim 35^\circ\text{C}$. Figure 6(a) shows the measured eye diagram for the 64-Gbit/s PAM-4 signal at the laser bias of 17 mA and the EAM bias of -0.82 V , where the input differential peak-to-peak voltage (V_{ppd}) at the RF pads of the evaluation board was 0.89 V . The average output power was -5.1 dBm . A fourth-order Bessel–Thomson filter and a 12-tap feed-forward equaliser (FFE) were applied. As shown in Fig. 6(a), a clear eye opening was achieved. The transmitter and dispersion eye closure quaternary (TDECQ) was 2.8 dB at a target symbol error ratio (SER) of 4.8×10^{-4} . The measured optical modulation amplitude (OMA) as a function of the input V_{ppd} is shown in Fig. 6(b). At an input V_{ppd} of 0.89 V , which was used for capturing the transmitter eye

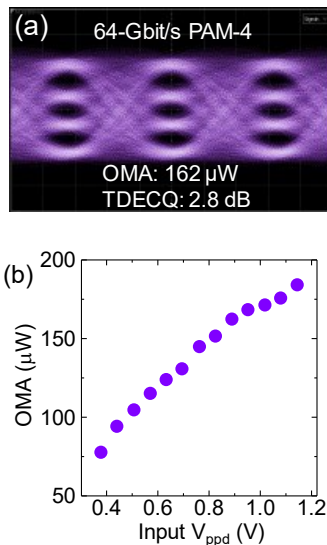


Fig. 6: (a) Optical eye diagram of the fabricated Tx chiplet at 64-Gbit/s PAM-4, measured using a reference Rx, and (b) OMA versus input differential V_{ppd} .

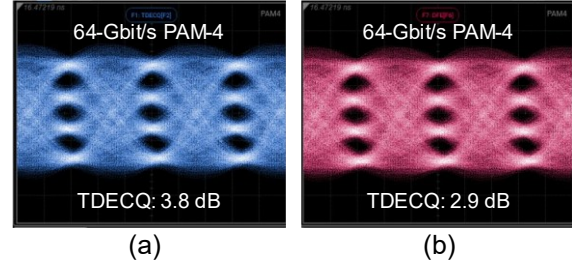


Fig. 7: Optical eye diagram of the chiplet-to-chiplet optical link using the fabricated Tx and Ge-PD-integrated Rx at 64-Gbit/s PAM-4 after (a) 28-tap FFE and (b) 28-tap FFE and 1-tap DFE.

diagram, the OMA and extinction ratio were $162\text{ }\mu\text{W}$ and 2.3 dB , respectively. The membrane DFB laser and the driver EIC consumed 36.2 and 40.4 mW , respectively. Thus, the energy cost of the Tx chiplet was 1.20 pJ/bit for 64-Gbit/s PAM-4 signals.

Finally, we examined chiplet-to-chiplet optical transmission for 64-Gbit/s PAM-4 signals by combining the Tx chiplet fabricated in this work with the previously reported Rx chiplet [13], as shown in Fig. 5. The Rx chiplet incorporates a Ge photodetector and a transimpedance amplifier (PD-TIA). The Tx and Rx chiplets were connected through 14 m of SSMF, with an optical isolator and a polarisation controller (PC) inserted in the link. The optical input power at the Rx fibre port was -4.9 dBm . Figures 7(a) and 7(b) show the eye diagrams measured at the output of the Rx chiplet at room temperature. In both cases, a fourth-order Bessel–Thomson filter was applied. The equalisation was performed using a 28-tap FFE for Fig. 7(a) and a 28-tap FFE followed by a 1-tap decision-feedback equaliser (DFE) for Fig. 7(b). Clear eye opening was observed in both cases, with TDECQ values of 3.8 and 2.9 dB , respectively, at the target SER of 4.8×10^{-4} . The PD-TIA consumed 22.3 mW , corresponding to an energy cost of 0.35 pJ/bit for 64-Gbit/s PAM-4 signals. As a result, the total energy cost of the Tx and Rx chiplets was $1.20 + 0.35 = 1.55\text{ pJ/bit}$. These results indicate that such low-power optical chiplets are promising for board-level interconnects in scale-up systems.

Conclusion

We fabricated an optical Tx chiplet with a heterogeneously integrated membrane InGaAsP DFB laser and a low-capacitance EAM on a Si photonics platform to investigate on-chip laser integration as an internal optical source. The Tx chiplet achieved 64-Gbit/s PAM-4 operation at an energy cost of 1.20 pJ/bit . When combined with a Ge-based Rx chiplet, it enabled a 1.55-pJ/bit optical link over 14 m of SSMF. These results show the potential of laser-integrated Tx chiplets as a low-power optical I/O solution for CPO systems.

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